GAI 601 — Machine Learning

Assignment 2: Exploratory Data Analysis (EDA)

Bank Marketing Dataset — Portuguese Bank Telemarketing Campaign

CUNY School of Professional Studies | Spring 2026

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Dataset: https://archive.ics.uci.edu/dataset/222/bank+marketing

Exploratory Data Analysis — Essay

Word count: **500 / 500 words (answers only, excluding question headings)**

Notation: Dataset column names — e.g. **duration**, **euribor3m** *— are shown in* **teal bold** *throughout for easy reference.*

The Bank Marketing dataset has 41,188 records and 21 features (columns/attributes). The target variable y predicts term deposit subscription (yes/no).

Q1. Are the Features Correlated?

Pearson correlation reveals multicollinearity: **euribor3m** correlates with **nr.employed** (r=0.945) and **emp.var.rate** (r=0.972). Client demographics show no high correlations. The strongest predictors are **nr.employed** (r=-0.355), **pdays** (r=-0.325), and **euribor3m** (r=-0.308), confirming subscriptions rise during economic downturns.

Q2. Is There Data Leakage?

The feature **duration** is a data leakage variable (r=+0.405), only known after a call ends when the outcome is determined. The UCI authors state **duration** should be discarded, as it causes overfitting and renders any model useless in deployment.

Q3. What Is the Overall Distribution of Each Variable?

The feature **age** is near-normal (mean=40). The features **duration** and **campaign** are right-skewed. The feature **pdays** is bimodal with 96.3% at value 999 (never contacted). Macroeconomic features

emp.var.rate, **euribor3m**, and **nr.employed** show two clusters reflecting pre- and post-2008 regimes.

Q4. Are There Any Outliers Present?

Using IQR, the feature **previous** has the highest outlier rate (13.7%), **duration** (7.2%) and **campaign** (5.8%). Macroeconomic features show zero outliers. Extreme **campaign** values should be Winsorized at the 99th percentile.

Q5. What Are the Relationships Between Different Variables?

Where **poutcome** equals success, clients subscribe at ~65% — six times the 11.3% average. The feature **contact** shows cellular outperforms telephone. March, September, and October far outperform may. Subscribers cluster in lower **nr.employed** and **euribor3m** periods.

Q6. How Are Categorical Variables Distributed?

The feature **job** is dominated by admin. (25.3%) and blue-collar (22.2%). The feature **default** has 20.9% unknown values. The feature **poutcome** is 86.3% nonexistent. The feature **month** shows may holding 33.4% of contacts but the lowest subscription rate.

Q7. Do Any Patterns or Trends Emerge in the Data?

The strongest pattern is that **poutcome** of success drives subscription to ~65%. Campaigns during lower **emp.var.rate** and **euribor3m** convert better. Rates decline after 3 **campaign** contacts. March, September, October, and December consistently outperform may despite may having the highest **contact** volume.

Q8. What Is the Central Tendency and Spread of Each Variable?

The feature **age** has mean=40, std=10.4. The feature **campaign** has mean=2.57, right-skewed. The feature **pdays** has 96.3% at 999. Macroeconomic features **emp.var.rate**, **cons.price.idx**, **cons.conf.idx**, **euribor3m**, and **nr.employed** show two distinct clusters reflecting different economic periods.

Q9. Are There Missing Values or Duplicates?

The dataset has no explicit NaN values. However, 12,718 implicit unknown values exist across: **default** (20.9%), **education** (4.2%), **housing** (2.4%), **loan** (2.4%), **job** (0.8%), and **marital** (0.2%). Twelve duplicates were identified for removal.

Q10. Recommendations for Data Cleaning and Modelling

Based on the full EDA, recommended actions are:

- Drop **duration** (data leakage, r=0.405); remove 12 duplicates.
- Impute unknowns using mode for **job**, **marital**, **housing**, and **loan**; treat **default** as three-category.
- Address severe class imbalance in y using SMOTE; evaluate with F1-score and AUC-ROC.
- Feature engineer previously_contacted from **pdays**, age_group from **age**, and econ_pressure from **euribor3m** × **emp.var.rate**.
- Drop **euribor3m**, **nr.employed**, and **emp.var.rate** — all replaced by the engineered econ_pressure feature.
- Focus campaigns on March, September, October, and December via cellular, maximum 3 attempts.

Code — Google Colab Notebook

The complete annotated Python code including all EDA outputs, visualizations, and analysis is available in the Google Colab notebook below. The notebook contains 15 cells covering all 10 EDA questions with detailed comments throughout.

To view the notebook:

- Click the link below — it opens directly in Google Colab
- All code outputs and charts are visible without running anything
- To re-run: Runtime → Run All (upload bank-additional-full.csv first)

Colab Notebook: https://colab.research.google.com/drive/1r1eucPtdTubSbUnJKR-b0j52gl1Rt7ho?usp=sharing

End of Assignment 2 — EDA | Sarosh Karim | GAI 601 | CUNY-SPS | 29-MAR-2026